"Osprey 3.5 V6: Low fuel rail pressure due to high-pressure fuel pump wear"

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Area: bulletins

1 Condition

Some Osprey 3.5 V6 vehicles may exhibit extended crank, power loss under heavy acceleration, and an illuminated MIL. Diagnostic scan stores DTC P0087, indicating fuel rail/system pressure too low, often accompanied by low live-data readings from the fuel rail pressure sensor FRS-4180. The concern is typically more pronounced when the fuel tank is below one-quarter and under sustained high demand. See Osprey Fuel Service Manual :: Fuel Pressure and Sensor Faults for the definition of P0087.

2 Cause

The root cause is mechanical wear of the high-pressure fuel pump HPFP-4120, where the cam-driven plunger and bore lose sealing efficiency and cannot maintain commanded rail pressure. As demanded pressure exceeds actual, the PCM sets DTC P0087. Inspect the pump and verify actual versus desired rail pressure per Osprey Fuel Service Manual :: High-Pressure Fuel Pump (GDI) and Osprey Fuel Service Manual :: Fuel System Diagnosis .

3 Repair Procedure

Replace the high-pressure fuel pump HPFP-4120 following procedure PROC-FUEL-PUMP as described in Osprey Fuel Service Manual :: Fuel Pump Replacement Procedure . Before condemning the pump, confirm the fuel rail pressure sensor FRS-4180 is reading accurately; if its signal is suspect, replace it per Osprey Fuel Service Manual :: Fuel Rail Pressure Sensor Replacement . Review Osprey Fuel Service Manual :: Fuel Delivery for torque and priming requirements. After repair, clear DTC P0087, prime the system, and verify rail pressure tracks the commanded value on a road test.